

Question 1

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How does Shor's algorithm threaten the security of RSA and Elliptic Curve Cryptography (ECC) and what are the potential consequences for current digital infrastructure?

Ans:

Shor's algorithm developed by mathematician Peter Shor in 1994, is a quantum algorithm capable of solving the integer factorization problem in polynomial time. RSA security is based on the computational difficulty of factoring large integers, and ECC relies on the hardness of the elliptic curve discrete logarithm problem. On classical computers, these

problems require exponential time for sufficiently large keys, making them secure in practice.

However, Shor's algorithm can solve both problems exponentially faster on a quantum computer, reducing the complexity from sub-exponential or exponential to polynomial time. This means that once sufficiently powerful quantum computers are available, they can efficiently break RSA and ECC encryption, rendering these systems obsolete.

Potential consequences include:

1. Data Breaches: Encrypted sensitive information such as banking data, health records, and classified government documents

could be decrypted.

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2. Digital Security Forgery: Attackers could forge digital certificates and signatures, compromising authentication mechanisms.

3. Retrospective Decryption: Encrypted communications intercepted and stored today could be decrypted in the future ("harvest now, decrypt later" attack)

4. Infrastructure Risk:

Critical systems such as financial transactions, secure communications and identity verification systems would require urgent migration to post-quantum cryptography.

Question 2

Discuss the role of quantum key distribution (QKD) in future cryptographic systems. How does it differ from classical public key encryption?

Ans:

QKD enables two parties to generate and share a secret key securely, with security guaranteed by quantum mechanics rather than computational hardness.

The most well-known protocol, BB84, uses the principle that any attempt to measure quantum states disturbs them, alerting parties to the presence of an eavesdropper.

Differences from classical public-key encryption:

1. Security basis: QKD relies on physical laws. PKI is based on mathematical problems.

2. Eavesdropping detection: In QKD interception alters the quantum state.

3. Quantum resistance: QKD is inherently secure against quantum computer attacks.

4. Implementation challenges:

QKD requires specialized hardware like single-photon emitters and detectors, making it harder to deploy.

Question 3

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What are the main differences between lattice based cryptography and traditional number-theoretic approaches like RSA, particularly in the context of quantum resistance?

Ans:

Differences:

1. Security foundation: Lattice problems are NP-hard, while RSA relies on factoring and ECC on discrete logs, both solvable by Shor's algorithm.
2. Quantum Resistance: Lattice-based schemes are post-quantum secure, making them candidates for future cryptographic standards.

3. Advanced capabilities: Lattice cryptography supports fully homomorphic encryption, identity-based encryption, and other advanced functionalities not easily achievable with RSA/ECC.

Question 4

Develop a python-based PRNG that uses the current system time and a custom seed value. Write complete program and corresponding output.

Example Python code:

```
import time
import random

# Create a custom seed using system
time and an arbitrary constant.
```



```
custom_seed = int(time.time()) 98754321
```

```
random.seed(custom_seed)
```

```
# Generate random numbers
```

```
numbers = [random.randint(0, 100) for _
```

```
in range(5)]
```

```
print("Seed used:", custom_seed)
```

```
print("Random numbers:", numbers)
```

Output:

Seed used : 1723719922

Random numbers : [23, 87, 5, 44, 91]

Question 5

Explain the Sieve of Eratosthenes algorithm and use it to find all

prime numbers less than 50. How does

its time complexity compare to

trial division?

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Ans:

Sieve of Eratosthenes algorithm:

1. Write down all numbers from 2 to n .

2. Start with $p=2$. Mark all multiples of p (except p) as composite.

3. Find the next unmarked number, set it as p , and mark all its multiples.

4. Repeat until $p > \sqrt{n}$

Example, $n=50$

Initial list: 2, 3, 4, ..., 50

After removing multiples of 2: 2, 3, 5, 7, 9, ...

continue for 3, 5, 7

Final primes < 50 : 2, 3, 5, 7, 11, 13, 17, 19,

23, 29, 31, 37, 41, 43, 47.

Time Complexity:

→ Sieve : $O(n \log n)$

→ Trial Division : $O(n\sqrt{n})$

Question 6

State and explain the necessary and sufficient conditions for a composite number to be a Carmichael number. Then verify whether the number $n=561$ and $n=1105$ and $n=1729$ are Carmichael numbers.

Ans:

A Carmichael number is a composite integer n that satisfies the congruence

$b^{n-1} \equiv 1 \pmod{n}$ for all integers b that are coprime to n .

These are Fermat pseudoprimes to all bases coprime to them.

Conditions:

→ n is square free (no repeated prime factors)

→ For every prime divisor p of n , $(p-1)$ divides $n-1$.

Verification:

→ $n = 561$: square free. ($3 \times 11 \times 17 = 561$)

→ check divisibility:

$p = 3 \rightarrow p-1 = 2$ divides $560 \rightarrow \text{yes}$

$p = 11 \rightarrow p-1 = 10$ " $560 \rightarrow \text{yes}$

$p = 17 \rightarrow p-1 = 16$ " $560 \rightarrow \text{yes}$

$\therefore 561$ is Carmichael.

Similarly,

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$$n = 1729 = 7 \times 13 \times 19$$

$$p = 7 \rightarrow p-1 = 6 \text{ divides } 1728 \rightarrow \text{yes}$$

$$p = 13 \rightarrow p-1 = 12 \text{ " " } \rightarrow \text{"}$$

$$p = 19 \rightarrow p-1 = 18 \text{ " " } \rightarrow \text{"}$$

$$n = 1105 = 5 \times 13 \times 17$$

$$p = 5 \rightarrow p-1 = 4 \text{ divides } 1104 \rightarrow \text{yes}$$

$$p = 13 \rightarrow p-1 = 12 \text{ " " } \rightarrow \text{"}$$

$$p = 17 \rightarrow p-1 = 16 \text{ " " } \rightarrow \text{"}$$

$\therefore 1729$ and 1105 are also Carmichael.

Question 7

Determine whether the following are valid algebraic structures and justify your answer: (a) Is it set $\mathbb{Z}_{11}$ with operations $(+, \cdot)$ a ring?

(b) Are the sets $(\mathbb{Z}_{37}, +)$ and $(\mathbb{Z}_{35}, \times)$ Abelian groups?

Ans:

(a) $\mathbb{Z}_{11}$ with $(+, \cdot)$ is a ring because:

→ $(\mathbb{Z}_{11}, +)$ is an Abelian group

→ Multiplication is associative,

→ Distributive laws hold.

→ Has identity 1.

(b) $(\mathbb{Z}_{37}, +)$ is an Abelian group (addition mod 37)

(c) $(\mathbb{Z}_{35}, \times)$ is not a group (many elements ~~don't~~ do not have inverses).

Question 9

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Determine the multiplicative inverse of 7 mod 26, if it exists. (Use extended Euclidean algorithm).

Ans:

We solve $7x \equiv 1 \pmod{26}$

Using Extended Euclidean Algorithm:

$$26 = 3 \times 7 + 5$$

$$7 = 1 \times 5 + 2$$

$$5 = 2 \times 2 + 1$$

Back substitute:

$$1 = 5 - 2 \times 2$$

$$= 5 - 2 \times (7 - 1 \times 5) = 3 \times 5 - 2 \times 7$$

$$= 3 \times (26 - 3 \times 7) - 2 \times 7$$

$$= 3 \times 26 - 11 \times 7$$

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$$\text{Thus } -11 \times 7 \equiv 1 \pmod{26} \rightarrow x \equiv -11 \equiv 15 \pmod{26}$$

Multiplicative inverse = 15.

Question 10

Evaluate $(-8 \times 5) \pmod{17}$ and explain how to simplify negative modular multiplication:

Ans:

First, compute the product: $(-8 \times 5) = -40$.

→ Reduce modulo 17:

$$-40 \div 17 = -3 \text{ remainder } 11$$

result 11.

Simplification:

First reduce negative numbers modulo n to get a positive equivalent.

then multiply.

Example: $-8 \bmod 17 = 9$. Then $(9 \times 5) \bmod$

$$17 = 45 \bmod 17 = 11$$

Question 11

State and prove Bezout's Theorem. Use it to find the multiplicative inverse of 92 modulo 385.

Ans:

Bezout's Theorem: For any integers a and b , there exist integers x and y such that $ax + by = \gcd(a, b)$.

Proof idea: The gcd can be expressed as a linear combination of a and b by the Extended Euclidean Algorithm.

To

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To find inverse of $97 \bmod 385$:

$\gcd(97, 385) = 1$, so inverse exists.

Using Extended Euclidean Algorithm:

$$385 = 3 \times 97 + 94$$

$$97 = 1 \times 94 + 3$$

$$94 = 31 \times 3 + 1$$

Back substitute:

$$1 = 94 - 31 \times 3$$

$$= 94 - 31(97 - 1 \times 94) = 32 \times 94 - 31 \times 97$$

$$= 32 \times (385 - 3 \times 97) - 31 \times 97$$

$$= 32 \times 385 - 127 \times 97$$

$$\text{Thus, } -127 \times 97 \equiv 1 \pmod{385} \rightarrow$$

$$\text{inverse} \equiv 258 \pmod{385}.$$

$$\text{Multiplicative inverse} = 258$$

Question 12

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Using Bezout's identity, prove the equation $ax + by = \gcd(a, b)$ has integer solutions. Find x such that $43x \equiv 1 \pmod{240}$.

Ans:

Bezout's identity states there are integers x, y such that $ax + by = \gcd(a, b)$

proof using extended Euclidean algorithm

For $43x \equiv 1 \pmod{240}$:

$$\gcd(43, 240) = 1 \rightarrow \text{inverse exists}$$

$$240 = 5 \times 43 + 25$$

$$43 = 1 \times 25 + 18$$

$$25 = 1 \times 18 + 7$$

$$18 = 2 \times 7 + 4$$

$$7 = 1 \times 4 + 3$$

$$4 = 1 \times 3 + 1$$

Back Substitute:

$$1 = 4 - 1 \times 3$$

$$= 4 - 1 \times (7 - 1 \times 4) = 2 \times 4 - 1 \times 7$$

$$= 2 \times (18 - 2 \times 7) - 1 \times 7$$

$$= 2 \times 18 - 5 \times 7$$

$$= 2 \times (43 - 1 \times 25) - 5 \times (25)$$

$$= 2 \times 43 - 7 \times 25$$

$$= 2 \times 43 - 7 \times (240 - 5 \times 43)$$

$$= 37 \times 43 - 7 \times 240$$

Thus, $37 \times 43 \equiv 1 \pmod{240}$

$$x = 37$$

Question 13

state Fermat's Little Theorem and explain how it is used to test for primality. Is 561 a prime number based on this test? Evaluate $5123 \bmod 175$ using Fermat's Little Theorem.

Ans:

Fermat's Little Theorem: If p is prime and $\gcd(a, p) = 1$, then $a^{(p-1)} \equiv 1 \pmod{p}$.

For Primality test: $a < n$, check if

$a^{(n-1)} \equiv 1 \pmod{n}$. If not, n is composite.

Test for $n = 561$ with $a = 2$:

$2^{560} \bmod 561 = 1$ (passes for some

a) but 561 is known composite.

so FLT can give false positives.

Evaluate $5123 \bmod 175$:

Since $175 = 25 \times 7$, use Chinese Remainder Theorem.

$$\text{Mod } 25: 5123 \equiv 0$$

$$\text{Mod } 7: 56 \equiv 1, \text{ so } 5123 \equiv 5$$

$$(6 \times 20 + 3) \equiv (120 \times 5 + 3) \equiv 125 \equiv 6 \bmod 7.$$

$$\text{CRT: } x \equiv 0 \bmod 25, x \equiv 6 \bmod 7$$

$$7 \rightarrow x = 125$$

$$\text{Answer} \equiv 125$$

Question 14

State and Prove the Chinese Remainder Theorem. Then solve $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 2 \pmod{7}$.

Ans:

Chinese Remainder Theorem (CRT):

if $n_1, n_2, \dots, n_k$ are pairwise coprime then the system of congruences has a unique solution modulo $N = n_1 n_2 \dots n_k$.

proof: constructing solution by successive substitutions or by explicit formula using modular inverses.

Solve:

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$$N = 3 \times 5 \times 7 = 105$$

$$N_1 = 35, N_2 = 21, N_3 = 15$$

Inverses:

$$35y \equiv 1 \pmod{3} \rightarrow y=2$$

$$21y \equiv 1 \pmod{5} \rightarrow y=1$$

$$15y \equiv 1 \pmod{7} \rightarrow y=1$$

Solution:

$$x = 2 \times 35 \times 2 + 3 \times 21 \times 1 + 2 \times 15 \times 1$$

$$= 140 + 63 + 30$$

$$= 233$$

$$233 \pmod{105} = 23$$

$$x \equiv 23 \pmod{105}$$

Question 15

Briefly explain the CIA triad in information security. How does each component contribute to building a secure system?

Ans:

CIA Triad: Confidentiality, Integrity, Availability.

→ Confidentiality: Ensuring information is accessible only to authorized parties, Achieved via encryption, access controls

→ Integrity: Ensuring data is accurate and unaltered. Achieved with hashing, digital signatures.

→ Availability: Ensuring systems/data are accessible when needed. Achieved via redundancy, backups, DOS protection.

All three must be balanced for a secure system.

Question 16

How does steganography differ from cryptography in the context of information security, and what are common techniques used for hiding data in digital media?

Ans:

cryptography: Protects data by transforming into unreadable form

without a key.

Steganography: Hides the existence of the message by embedding it in other data (e.g. images, audio).

Common techniques:

→ LSB (Least Significant Bit)

→ Audio echo hiding

→ Video frame manipulation

Question 17

What are the key differences between phishing, malware and denial-of-service (DOS) attacks in terms of their method and impact on system security?

Ans.

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→ Phishing: Social engineering attack to trick users into revealing credentials. Delivered via email, fake sites.

→ Malware: Malicious software that disrupts, damages or gains unauthorized access (e.g. viruses, ransomware)

→ DOS: Overwhelms a service with traffic, rendering it unavailable.

Question 18

Explain how legal frameworks such as the General Data Protection

Regulation (GDPR) help mitigate cyber attacks and protect user privacy.

Ans:

GDPR enforces strict rules on data collection, processing and storage:

→ Requires consent for data collection.

→ Right to access, correct, delete personal data.

→ Mandatory breach notification within 72 hours.

→ Heavy penalties for non-compliance.

Explain the basic working of the DES algorithm using a simple 64-bit plaintext block and a 56-bit key.

Ans:

DES (Data Encryption Standard) is a symmetric block cipher:

1. Initial permutation (IP) of plaintext.
2. Split into L_0 and R_0 (32 bits each)
3. 16 rounds : $R_i = L_{i-1} \text{ XOR } f(R_{i-1}, K_i)$, $L_i = R_{i-1}$
4. f -function : Expansion (32 $\rightarrow$ 48 bits)
XOR with K_i : S-box.
5. Final swap and inverse IP.

Key schedule generates 16 subkeys from the 56-bit key.

Question 20

In the DES algorithm, given $L_0 = 0$
 $0xFOFOFOFO$, $R_0 = 0x0F0F0F0F$, round key
 $K_1 = 0x0F0F0F0F$, compute first round
 output.

Ans:

$$f(R_0, K_1) = R_0 \oplus K_1 = 0x0F0F0F0F \\ \oplus 0x0F0F0F0F = 0x00000000$$

Then $L_1 = R_0 = 0x0F0F0F0F$,

$$R_1 = L_0 \oplus f(R_0, K_1) = 0xFOFOFOFO \oplus \\ 0x00000000 = 0xFOFOFOFO$$

Question 21

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Given the input word [0x23, 0x47, 0x4C, 0x19]. Use the following partial AES S-box to perform the SubBytes transformation. Provide the resulting output word.

Row ↓ / Col →	3	4	7	9	A	C
1	6D	.	.	C6	.	.
2	D4	.	.	.	.	.
4	.	A1	.	.	.	2E
A	.	.	63	.	D2	.

Ans:

Applying AES S-box using standard values:

P.T.O.

$$S(0x23) = 0x26$$

$$S(0xA7) = 0x5C$$

$$S(0x4C) = 0x29$$

$$S(0x19) = 0xD4$$

Output word after subBytes [0x26,
0x5C, 0x29, 0xD4]

Question 22

In AES encryption, apply the AddRound-Key step only. Given: Input word;

[0x1A, 0x2B, 0x3C, 0x4D]; Round

key word: [0x55, 0x66, 0x77, 0x88]

Perform X-OR between the input word and the round key. and write the resulting output word.

Ans:

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$$0x1A \oplus 0x55 = 0x4F$$

$$0x2B \oplus 0x66 = 0x4D$$

$$0x3C \oplus 0x77 = 0x4B$$

$$0x4D \oplus 0x88 = 0xC5$$

Output: $[0x4F, 0x4D, 0x4B, 0xC5]$

Question 23

Show how AES Mixcolumns uses the fixed matrix over $GF(2)$:

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix}$$

to transform an input column.

Use an example column with values

$$[0x01, 0x02, 0x03, 0x04]$$

Ans:

For input column $a = [a_0, a_1, a_2, a_3] =$

$[01, 02, 03, 04]$, the output r

is:

$$r_0 = (0 \cdot 2 \cdot a_0) \oplus (03 \cdot a_1) \oplus (01 \cdot a_2) \oplus (01 \cdot 03)$$

$$r_1 = (01 \cdot a_0) \oplus (02 \cdot a_1) \oplus (03 \cdot a_2) \oplus (01 \cdot a_3)$$

$$r_2 = (01 \cdot a_0) \oplus (01 \cdot a_1) \oplus (02 \cdot a_2) \oplus (03 \cdot a_3)$$

$$r_3 = (03 \cdot a_0) \oplus (01 \cdot a_1) \oplus (01 \cdot a_2) \oplus (02 \cdot a_3)$$

$$\begin{aligned} \therefore r_0 &= 0x03, r_1 = 0x04, r_2 = 0x09, \\ r_3 &= 0x0A \end{aligned}$$

$$\therefore ([01, 02, 03, 04]) = [0x03, 0x04, 0x09, 0x0A]$$

Question 24

Describe how AES-OFB mode works.
How does it ensure synchronization between encryption and decryption streams?

Ans:

How it works:

→ It begins with an initialization vector (IV)

→ The IV is encrypted with AES to produce a keystream block.

→ This keystream is XORed with the plaintext to generate ciphertext.

→ For the next step, the AES output is fed back into AES to generate the next keystream block.

→ This process repeats until all plaintext is encrypted.

Synchronization:

Encryption and Decryption both generate the same keystream sequence from the same IV. As long as sender and receiver start with the same IV and remain

in step, their keystreams match, so ciphertext X-OR keystream recovers the original plaintext.

Question 25

Which AES modes cause error propagation during encryption, and how does this affect the integrity of the decrypted message?

Illustrate with CBC and CFB mode.

Ans:

→ ECB, CTR, OFB: No error propagation.

→ CBC: One corrupted ciphertext block corrupts the current plaintext block.

→ CFB: An error affects the current block.

1. CTR mode:

→ Decryption formula: $P = AES^{-1}(C_1 \oplus C_1^{-1})$

→ Error effect:

→ if c_i has a 1-bit error → P_1 becomes completely random.

2. CFB

→ Decryption formula: $P_i = C_i \oplus AES(C_{i-1})$

→ Error effect: - P_{i+1} becomes random.

Question 26

Which AES mode would you recommend for encrypting large files with parallel processing, and why?

Ans:

CTR is much more recommended for encrypting large files with parallel processing.

Reasons:

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→ Parallelism: CTR encrypts independent counters that can be processed in parallel, giving excellent outputs.

→ Random access: Supports seeking/decrypting any block independently, ideal for large files.

→ Security: Avoids ECB's pattern leakage. Unlike CBC, which chains blocks

→ CTR blocks are independent.

P.T.O.

Question 27

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Given a message "A" represented as $M=1$. encrypt it using public key $e=5$, $n=14$, what is the ciphertext? Then decrypt using private key $d=11$.

Ans:

RSA encryption: $C = M^e \bmod n = 1^5 \bmod 14 = 1$

Decryption: $M = C^d \bmod n = 1^{11} \bmod 14 = 1$

Ciphertext = 1.

Recovered plaintext = 1.

Question 28

Given : Message hash $H(M) = 5$, RSA private key : $d = 3$, $n = 33$, Generate the digital signature.

Ans:

$$\text{RSA signature } s = H(M)^d \bmod n = 5$$

$$3 \bmod 33 = 125 \bmod 33$$

$$33 \times 3 = 99 ; 125 - 99 = 26$$

Signature = 26 (Verification uses n .

$e \equiv H(M) \pmod{n}$ with the public exponent e)

p-T.O.

Question 29

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Aleya and Badol are using the Diffie-Hellman key exchange protocol. They agree on the following public values: Prime $p=23$; Base (generator) $g=3$. Aleya chooses private key $a=4$.

Badol chooses private key $b=5$. Compute the public keys and shared secret key.

Ans:

Public keys:

$$\text{Aleya: } A = g^a \text{ mod } p = 3^4 = 81 \equiv 12 \pmod{23}$$

$$\text{Badol: } B = g^b \text{ mod } p = 3^5 = 243 \equiv 13 \pmod{23}$$

Shared Secret:

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$$k = B \cdot a \bmod p = 13$$

$$4 \bmod 23$$

$$13^2 \equiv 169 \equiv 8 \pmod{23};$$

$$13^4 \equiv 8, \quad 2 = 64 \equiv 18 \pmod{23}$$

(Equivalent via $A \cdot b \bmod p = 12, 5 \equiv 18$)

So, $A=12, B=13$. shared key $k=18$

Question 30

A simple (non-cryptographic) hash function $H(x)$ is defined as the sum of ASCII values of the characters in x , modulo 100: $H(x) = (\sum \text{ASCII}(x_i))$

mod 100. Compute $H("AB")$ and $H("BA")$. Do the two messages produce the same

hash? What does this imply about collision resistance?

Ans:

ASCII: 'A' = 65, 'B' = 66.

$$H('AB') = (65 + 66) \bmod 100 = 131 \bmod 100 = 31$$

$$H("BA") = (66 + 65) \bmod 100 = 131 \bmod 100 = 31$$

Yes, both produce the same hash.

∴ The function is not collision resistant.

Question 31

In a secure message system, a simple message Authentication code

(MAC) is computed as $MAC = (Message + SecretKey) \bmod 17$. Given: Message = 15, SecretKey = 7. Compute the MAC.

Suppose an attacker changes the message to 10 but does not know the key. Can they forge the correct MAC easily? Explain briefly.

Ans:

$$MAC = (15 + 7) \bmod 17 = 22 \bmod 17 = 5$$

If the attacker observes $(M, MAC) = (15, 5)$, they can compute the

key:

$$SecretKey \equiv MAC - M \equiv 5 - 15 \equiv -10 \equiv 7 \pmod{17}$$

Then they can forge MAC for $M \neq 10$ as $(10+x) \bmod 17 = 0$.

Therefore, this MAC is not secure.

Real MACs avoid this weakness.

Question 32

Explain the steps involved in the TLS handshake process. How are symmetric keys established securely using asymmetric cryptography during the handshake?

Ans:

Typical TLS 1.2 (ECDHE) handshake flow:

1. ClientHello.
2. ServerHello

3. Key Exchange:

4. Key derivation

5. Finished messages.

Security establishment:

→ Security comes from authenticated

ECDFE providing forward secrecy

and resistance to passive eavendropping

Asymmetric crypto authenticates the

server; symmetric keys are

derived from EDCE shared secret.

Question 33

Explain the layered architecture

(protocol stack) of SSH. Briefly describe

the roles of each layer.

Ans:

SSH has three logical layers:

1. Transport layer (ssh-transport):

Negotiates algorithms, performs key exchange, provides server authentication, and sets up encryption/MAC for confidentiality and integrity.

2. User authentication Layer (ssh-userauth):-

Authenticates the client user via passwords, public keys, or keyboard-interactive methods.

3. Connection layer (ssh-connection):

Multiplexes the encrypted tunnel into multiple logical channels.

Question 34

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Explain the TLS handshake again, focusing on message flow and security properties (freshness, server authentication, and forward secrecy)

Ans:

Message Flow (ECDHE based)

ClientHello $\rightarrow$ ServerHello,

Certificate (Certificate Status/Ocsp),

Server Key Exchange (ECDHE params),

(Certificate Request), Server Hello Done

$\rightarrow$ Client key exchange (Client's

ephemeral key), (Certificate,

Certificate Verify), Change Cipher Spec

Finished $\rightarrow$ Server Cipher Spec, Finished.

Security Properties:

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→ Perfectness

→ Server authentication

→ Forward secrecy

→ Integrity and confidentiality.

Question 35

What is the general form of an elliptic curve equation over a finite field, and why is it used in cryptography?

Ans:

Over a prime field F_p ($p > 3$),

the short Weierstrass form is:

$$y^2 \equiv x^3 + ax + b \pmod{p}, \text{ with}$$

$$\text{discriminant } \Delta = -16(4a^3 + 27b^2) \neq 0$$

to avoid singularities.

The set of points (x, y) satisfying the equation plus the point at infinity forms an Abelian group under a geometric chord-and-tangent addition law.

Reason to use:

The elliptic curve Discrete logarithm problem (ECDLP) $\rightarrow$ given P and $Q = kP$, find k is believed to be hard, allowing small key sizes with high security. ECC provides efficient key exchange (ECDH) signatures (ECDSA/EdDSA) and performance advantages, especially on constrained devices.

Question 36

How does ECC achieve the same level of security as RSA with a smaller key size? Briefly explain.

Ans:

ECC's security is based on the ECDLP, for which the best-known classical attacks (e.g.: Pollard's rho) have exponential complexity in $\sqrt{n}$, where n is the group order. In contrast, RSA's hardness (integer factorization) has sub-exponential attacks. Consequently, ECC can use much smaller keys to achieve comparable security. Smaller keys mean faster computations, lower

bandwidth, and reduced storage, benefitting many applications including mobile and IoT.

Question 37

Given the elliptic curve $y_2 \equiv x_3 + 2x + 3 \pmod{97}$ determine whether the point $p = (3, 6)$ lies on the curve.

Ans:

compute both sides modulo 97:

$$\text{LHS: } y_2 \equiv 6 \equiv 36 \pmod{97}$$

$$\begin{aligned} \text{RHS: } x_3 + 2x + 3 &\equiv 3^3 + 2 \cdot 3 + 3 = 27 + 6 + 3 \\ &\equiv 36 \pmod{97} \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS}$$

$\therefore p$ lies on the curve.

Question 38

Given public key $(p=23, g=5, h=8)$ and message $m=10$, compute the ElGamal ciphertext using random $k=6$.

Ans:

ElGamal encryption outputs (c_1, c_2) :

$$c_1 = g^k \bmod p = 5^6 \bmod 23, 5^2 = 25 \equiv 2;$$

$$5^4 \equiv 2^2 = 4; 5^6 \equiv 4 \times 2 = 8 \pmod{23}$$

$$\Rightarrow c_1 = 8.$$

$$c_2 = m \cdot h^k \bmod p.$$

$$h^k = 8^6 \bmod 23;$$

$$8^2 = 64 \equiv 18; 8^4 \equiv 18^2 = 324 \equiv 2;$$

$$8^6 \equiv 8 \cdot 4 \times 8$$

$$2 \equiv 2 \times 18 = 36 \equiv 13 \pmod{23}$$

$$\text{Therefore } c_2 = 10 \times 13 \equiv 130 \equiv 15 \pmod{23}$$

Ciphertext : $(C_1, C_2) = (8, 15)$

Question 39

Explain how lightweight cryptography is important for securing IOT devices. Give one example of a lightweight encryption algorithm used in IOT.

Ans:

IOT devices often have limited CPU, memory power and bandwidth. Lightweight cryptography delivers security with small code size, low RAM usage, and minimal energy consumption while maintaining acceptable latency. It enables secure boot, firmware

updates, device-to-cloud communication.

Example algorithm:

→ ASCON

→ PRESENT, GIFT

→ Speck/Simon

→ CLEFIA

ASCON is commonly used.

Question 40

List and briefly explain the common IOT-specific attacks (e.g.: firmware hijacking physical tampering, botnets like Mirai). What mitigation strategies can be applied?

P-T-O.

Common IOT attacks:

- Firmware hijacking
- Supply-chain attacks
- Physical tampering
- Botnet recruitment
- ~~Insecure update mechanisms~~
- Why these happen:
 - Insecure update mechanisms
 - Insecure communications
 - Inadequate isolation:

Mitigation:

- Strong authentications
- Secure boot and signed updates
- Hardening and least privileges
- Physical security
- Network defenses.
- Monitoring and asset management
- Lifecycle management.